STRUCTURAL DYNAMICS RESEARCH

Traffic Organization and Bridge Deck Fatigue

Numerical fatigue performance under traffic loading

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Research question

How does traffic organization influence concrete deck fatigue performance under the same vehicle fleet?

Computational model

26,520

MITC4 bending/shear plate elements

Coupled steel beams and vertical supports
Approximately 79,600 algebraic coordinates
Linear equivalent model in MATLAB

Study design

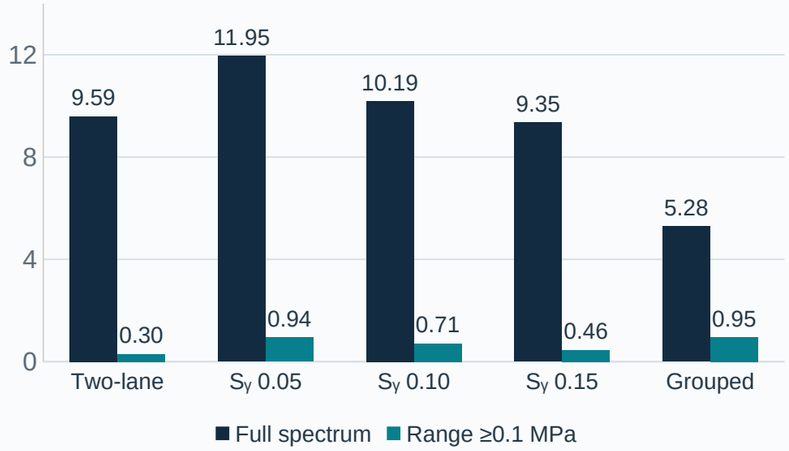
22 selected traffic scenarios use the same 240-vehicle fleet, including 20 heavy vehicles.

The comparison varies longitudinal spacing, lateral spread and vehicle ordering, with two-lane and grouped-traffic references.

Compression fatigue

Cumulative damage $D = \Sigma\, n_i/N_i$ uses the adopted concrete compression S-N relationship. The same eligible observation locations define the comparison.

Maximum damage per 240 vehicles ($\times 10^{-14}$)



d = 40 m, traffic increments, K = 17. The range cutoff tests sensitivity to small cycles. Tensile-exceedance locations remain outside the compression assessment.

3 locations / 7 episodes

Grouped traffic exceeds the tensile threshold at three common observation points.

Stress-range coverage

Nominal sections cover all saved observation coordinates.

Recorded stress ranges reach at least 98.53% of the interior field-range upper bound. This measures range-to-bound coverage.

Numerical checks

3,808 independent section-resultant checks and 23,680 cycle/parameter recomputations verify the implementation.

The maximum relative error in recalculating damage is 8.6×10^{-15}.

Fatigue interpretation

n_i: counted cycles
N_i: cycles from the adopted S-N model

Controlling damage uses S-N continuation beyond two million cycles. Small-cycle treatment affects comparisons. Tensile exceedance remains a separate outcome.

Individual work: model reconstruction, implementation, verification, simulation review and documentation. Professor Kai-Chun Chang supplied the bridge drawings.

Conditional numerical fatigue assessment. Loading, S-N continuation and cycle-cutoff assumptions limit interpretation. No actual service-life prediction.